

Algorithm:

1. Cones are divided into two lists to identify the sides of the track:

- a. Blue: Left side cones (color==1)
- b. Yellow: Right side cones (color==0)

2. Avg(points):

A function that calculates the average position of the cones on each side to follow the center if the track:

- If there are 0 cones, returns none
- If there is 1 cone, return its position
- If there are 2 cones, return the midpoint between them
- If there are 3 cones (added for part 2 of the task), return the average of the 3 positions

So, the codes can handle up to 3 cones.

3. Handling Cone cases:

Case	Condition and Action
1	If both sides have cones, the center pt is the midpoint between the yellow and blue positions
2	If only one cone exists on one side, offset the path 1 meter away from that cone either left or right depending on the cone color and where it is positioned
3	If no cones are there, go straight ahead

4. Generate the path

Once the target direction is known which is the center of both x and y components

- The angle is computed
- The car moves forward in steps (0.5 m)

Note:

The current algorithm implemented only moves in straight line toward the computed center, still configuring the turning and curve handling so a few test cases failed, but will be configured soon inshalaah :)